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Mining for sand extraction - Rio Grande: Brazilian environmental legislation applied

Mineração para extração de areia - Rio Grande: legislação ambiental brasileira aplicada

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Abstract

Sand is a fundamental natural resource, and its availability is crucial for economic growth and infrastructure development. The present work aimed to analyze the sand mining processes in an enterprise on the banks of the Rio Grande, in the southwest of Minas Gerais, mainly the environmental aspects and the focus on meeting legal conditions and requirements that encompass the process of extracting sand. sand extraction is a globally widespread activity, essential in the construction industry and several other industrial applications. Rivers provide a natural source of sand, a vital component in manufacturing concrete, asphalt, glass, and ceramic products and foundry. However, this activity is not without controversy, due to the significant environmental impacts it entails. In extracting sand from the bed of rivers or lakes, a dredger is used to remove sand and other materials found at the bottom of the water body, using water as a means of transport. Sand extraction is a globally widespread activity, playing an essential role in the construction industry and several other industrial applications. Rivers provide a natural source of sand, a vital component in manufacturing concrete, asphalt, glass, and ceramic products and foundry. However, this activity is not without controversy, due to the significant environmental impacts it entails. In extracting sand from the bed of rivers or lakes, a dredger is used to remove sand and other materials found at the bottom of the water body, using water as a means of transport. The main operation of this project consists of the extraction of sand and gravel intended for direct use in civil construction, as established by code A-03-01-8 in COPAM Normative Deliberation no. 217/2017 (Brazilian State Legislation), being a potential polluting activity. Furthermore, the aforementioned development is located within the buffer zone of the Serra da Canastra National Park, weighing 1 for the locational framing criteria.

Keywords: Mining. Water Resources. Environmental Management.

Resumo

A areia é um recurso natural fundamental, e sua disponibilidade é crucial para o crescimento econômico e o desenvolvimento de infraestruturas. O presente trabalho teve como objetivo analisar os processos de mineração de areia em um empreendimento as margens do Rio Grande, no sudoeste de Minas Gerais foram analisados principalmente os aspectos ambientais e o foco no atendimento de condicionantes e requisitos legais que englobam o processo de extração de areia, A prática da extração de areia é uma atividade globalmente difundida, desempenhando um papel essencial na indústria da construção civil e em diversas outras aplicações industriais. Os rios fornecem uma fonte natural de areia, um componente vital na fabricação de concreto, asfalto, vidro, bem como na produção de produtos cerâmicos e fundição. No entanto, essa atividade não é isenta de polêmicas, devido aos significativos impactos ambientais que acarreta. Na operação de extração de areia a partir do leito de rios ou lagos, uma draga é empregada para retirar a areia e outros materiais que se encontram no fundo da massa d'água, utilizando a água como meio de transporte. A prática da extração de areia é uma atividade globalmente difundida, desempenhando um papel essencial na indústria da construção civil e em diversas outras aplicações industriais. Os rios fornecem uma fonte natural de areia, um componente vital na fabricação de concreto, asfalto, vidro, bem como na produção de produtos cerâmicos e fundição. No entanto, essa atividade não é isenta de polêmicas, devido aos significativos impactos ambientais que acarreta. Na operação de extração de areia a partir do leito de rios ou lagos, uma draga é empregada para retirar a areia e outros materiais que se encontram no fundo da massa d'água, utilizando a água como meio de transporte. A principal operação desse empreendimento consiste na extração de areia e cascalho destinados à utilização direta na construção civil, conforme estabelecido pelo código A-03-01-8 na Deliberação Normativa da COPAM no 217/2017, sendo uma atividade de potencial poluidor M. Além disso, o referido empreendimento está localizado dentro da zona de amortecimento do Parque Nacional Serra da Canastra, tendo peso 1 para os critérios locacionais de enquadramento.

Palavras-chave: Mineração. Recursos Hídricos. Gestão Ambiental.

Introduction

Sand is an abundant natural resource, widely used in the manufacture of products vital to our daily lives. However, this activity is not without significant controversy and challenges, as sand extraction can have profound impacts on the environment and society.

The present work aimed to carry out an analysis from the perspective of an enterprise that extracts sand in the southeast of Minas Gerais, the activity carried out is characterized as anthropic, that is, human intervention, as the vast majority of enterprises cause a direct impact on the environment. environment, however, is fundamental for the development of society with the inputs provided by the activity. Brazil stands out for the production of raw sand sold, which in 2005 reached a total of 141,084,561 m^3, around R$ 1,923,468,414 in profit**. (MARQUES 2010).**

In summary, sand extraction plays a key role in construction and economic development, but the environmental and social challenges associated with this activity require a careful and sustainable approach. It is important to balance human needs with preserving ecosystems and implementing responsible sand mining practices.

It is an important sector for the economic development of the region, it is considered an activity of social interest for the population since it is necessary to use the extracted resources for applications in various sectors of society (**ANNIBELLI & MARÉS 2006).**

According to BRAND 1998, sand mining activities have a significant impact on economic and social development, however, they can cause environmental impacts and irreversible damage. The sub-basins where the projects are installed are those that are directly impacted by the activities.

Mining activity is one of those that most affect the surface, and it is not just the place where the project is located that suffers from changes in the soil; water, and air (GRIFFITH,1980).

Generally, sand mining is classified as the extraction of sand and gravel for immediate use in civil construction. - A-03-01-8" in Normative Deliberation 217 - State Council for Environmental Policy - COPAM, are included in Permanent Protection Areas (APP) Law No. 12,651, of May 25, 2012, which will provide for the guidelines and standards for the preservation of the banks of water bodies. According to State Decree 47,749/2019, in Art. 17:

> "Environmental intervention in APPs may only be authorized in cases of public utility, social interest and occasional activities or low environmental impact, and the lack of a technical and locational alternative must be proven."

The activity of sand extraction is of social interest under the Forestry Law of the State of Minas Gerais nº 20,922/2013, where it cites in its 3rd article the following:

> "II – Social interest f) research and extraction activities for sand, clay, gravel, and gravel, granted by the competent authority;

The visited enterprise carried out the intervention in APP as presented in the laws above as it was used for social interest (FIGURE 1). It is located in the Atlantic Forest biome. If there is a need for "2 to 1" compensation, "2 to 1" compensation refers to an environmental compensation strategy that requires the restoration or preservation of an area equivalent to twice the impacted area. This means that, if a certain human activity causes an environmental impact in an area of the Atlantic Forest, those responsible for the activity must guarantee the preservation or restoration of twice that area in another location to compensate for the damage caused. This type of compensation is a more rigorous measure that aims to ensure effective compensation for negative impacts on natural areas. Mainly governed by Federal Law No. 11,428/2006, known as the Atlantic Forest Law. **Figure 1: Permanent Protection Area.**

Figure 1: Permanent Protection Area.
Source: Google Earth 2023.

It is essential to ensure that sand mining is carried out responsibly, thus minimizing negative impacts and maximizing benefits for all parties involved and once explored, the Federal Constitution of 1988 provides, in its article 225, §2, that

> "Anyone who exploits mineral resources is obliged to recover the degraded environment, by the technical solution required by the competent body, by the law".

This is an important provision to guarantee environmental preservation and the recovery of areas impacted by mineral exploration. The venture visited is located between the large bodies of water between Passos and São João Batista do Glória. The Rio Grande is a river of great importance in the region of São João Batista do Glória. It rises in Serra da

Canastra, an environmental preservation area known for its natural beauty and rich biodiversity.

The river flows through several municipalities, including São João Batista do Glória, before emptying into the Paraná River. The Rio Grande is essential for water supply, agriculture, and fishing in the region. Furthermore, its scenic beauty attracts tourists and nature lovers to practice outdoor activities, such as ecotourism and adventure tourism.

Figure 2: Representation of Rio Grande.
Source: IDESISEMA data.

Material and Methods

The company has a dedicated diesel storage facility, in a certified storage tank, designed to meet safety standards and environmental regulations, and kept within a covered area designed to ensure safety. The diesel storage area is equipped with a containment system with channels around it, designed to capture and contain any possible fuel leaks, as shown in Figure 3.

Figure 3: Diesel storage tank.
Source: The authors 2023.

An essential part of the company's commitment to sustainability is the implementation of an effective reverse logistics system. This means that, after using diesel packaging, the enterprise assumes the responsibility for collecting, recycling, or disposing of these packaging in a safe and environmentally responsible way, contributing to the reduction of waste and the sustainable use of natural resources, thus promoting a more efficient and ecological cycle in the management of diesel packaging. Sand extraction takes place on the banks of the Rio Grande, involving the removal of ore (sand) in its natural state, with a simple classification step to remove organic materials present in the sand.

In addition to providing high-quality sand for construction and other industries, the enterprise seeks to diversify its sources of revenue by exploring the ornamental potential of the extracted material that is not used for civil construction, as shown in Figure 4.

Figure 4: Material not used in extraction.
Source: The authors 2023

During the mining process, these materials are removed along with the sand and those that meet beauty and durability standards can be separated to be later worked and transformed into garden decorations. This includes cutting, polishing, and shaping stones to create unique and attractive pieces that can be used in landscaping and gardening.

The extraction process, Figure 5, is carried out using a barge equipped with a dredger, which moves to the extraction site, respecting the limits of its designated area. Upon reaching the extraction point, the barge lowers the anchors, activates a winch, and releases the cables that support the pipeline. After stabilization on the sandbank, the diesel engine is activated to vacuum the sand, along with the water, which is transported to the barge's sandbox.

Figure 5: Barge equipped with a dredger.
Source: The authors 2023.

The barge arrives at the unloading area, where a team of qualified operators is ready to conduct the unloading process efficiently and safely. The barge is firmly anchored, ensuring its stability during operation. Subsequently, the unloading process begins with the activation of the machinery that controls the piping used to transport the sand from the barge to the screening area, Figure 6. The end of the piping is positioned at the designated dump site, where the sand will be processed. When starting the engine, the suction process begins, and the mixture of water and sand is transported through the pipes to the sieving area.

Figure 6: Unloading.
Source: The authors 2023.

The sand is dumped onto a platform that takes it to the sieve. Sieving, Figure 7, allows you to separate the sand according to the desired specifications, ensuring that only particles of the correct size go through the process, while unwanted particles are discarded in specific containers, Figure 8.

Figure 7: Sieving the material.
Source: The authors 2023.

Figure 8: Deposit box for unused material.
Source: The authors 2023.

Control of the entire process is carried out by trained operators who closely monitor the unloading rate, sand quality, and operational safety. They are also prepared to intervene in the event of any problems, ensuring a continuous flow of material for the next stage of the aggregate production process.

Barge unloading is a crucial operation, ensuring that sand extracted from the river is prepared and processed effectively and by customer specifications, contributing to the production of high-quality construction materials. Subsequently, this sand is transported to the storage point, Figure 9. The extraction method involves the phases of exploration, transportation, storage, loading, and commercialization

Figure 9: Storage point.
Source: The authors 2023.

At the mining site, a water management system is implemented to ensure that water used during sand extraction is returned to the source river cleanly and safely. This system consists of a network of channels strategically positioned to collect the water used in various stages of the extraction process and send it back to the river, Figure 10.

Figure 10: Channels for water drainage.
Source: The authors 2023.

The channels are designed with gentle slopes and carefully planned paths to allow water to flow naturally and be collected at meeting points. These gathering points, Figure 11, help separate sediment and other suspended materials from the water before returning to the river. This ensures that the water returned is as clean as possible.

Figure 11: Box for sedimentation of particulate matter.
Source: The authors 2023.

Results and Discussion

Sand extraction is a globally widespread activity, playing an essential role in the construction industry and several other industrial applications.

According to Tobias et al. (2010), the sand extraction operation in the submerged bed of rivers and lakes triggers a series of significant impacts. These include increased turbidity in the water, modification of the original course of the water body, potential interference with the hydrodynamic dynamics of lakes or rivers, water contamination due to the use of oils and greases on vessels, the elimination of areas of refuge for aquatic fauna, visual disturbance and increased risk of accidents due to boat traffic.

However, positive impacts are also observed, such as the reduction of siltation in water courses and the increase in sand availability in the region. Therefore, it is important to understand and face these challenges to ensure that sand extraction in rivers is conducted sustainably and responsibly, reconciling the needs of the industry with environmental preservation.

Conclusions

The sand mining operation in a river is a complex activity that involves several environmental, social, and economic aspects. Sustainability is the fundamental pillar for the long-term success of this venture. From an environmental point of view, the enterprise operates with maximum respect for the river's ecosystem, adopting responsible extraction practices, mitigating negative impacts, and implementing water and land management systems. Preserving the aquatic environment, maintaining water quality, and protecting biodiversity are essential priorities.

In the social context, there is concern on the part of managers with the safety of workers, the prevention of conflicts with local communities, and the promotion of sustainable economic development in the region, including the creation of jobs, support for community projects, and respect for local traditions and cultures. In economic terms, sand mining can significantly contribute to the region's development, providing essential materials for civil construction and other industries. Furthermore, the diversification of revenue sources, such as the sale of derivative products, such as gardening ornaments, can boost the economic growth of the enterprise. Last but not least, compliance with current legislation is of critical importance.

Respecting environmental, labor, and other regulations is essential to avoid fines, ensure long-term sustainability, and build a reputation for corporate responsibility. In summary, sand mining in a river can be a beneficial activity if carried out responsibly, with respect for the environment, social commitment, and in compliance with legislation. The search for a balance between environmental, social, and economic aspects is fundamental to the lasting success of this enterprise.

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